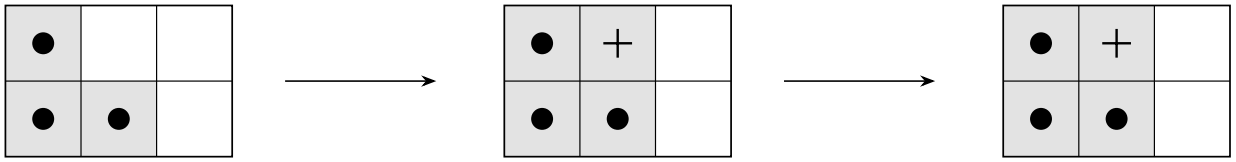
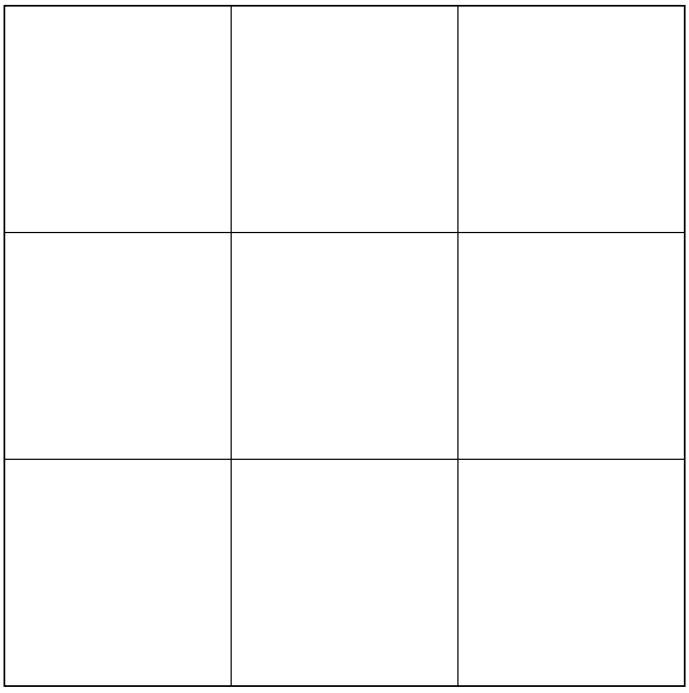
Week 98 / Growing from a few seeds / Grades 4–5

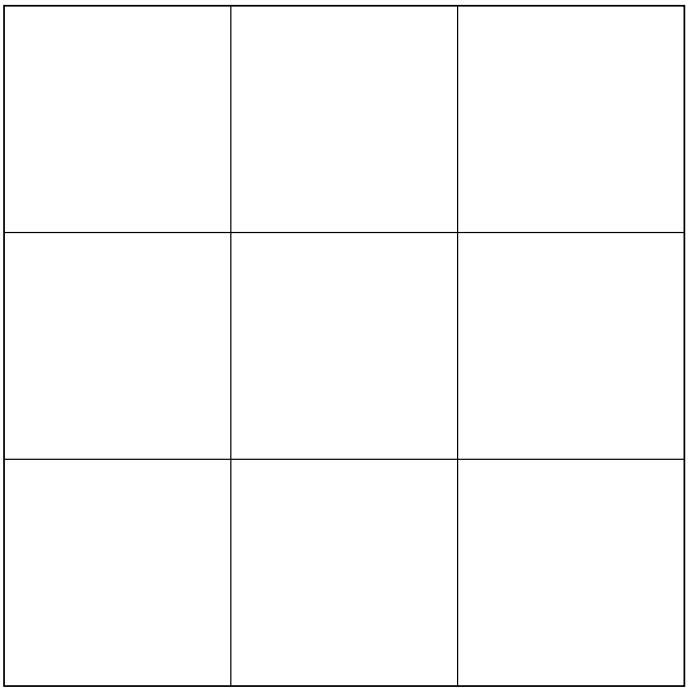
Seeds are the squares filled at the start. Put counters on them. Add a counter to an empty square when at least two squares sharing its sides are filled. Your partner points to two filled neighbors before each addition. Counters stay put. Stop when no more can be added. Diagonal touches do not count. There are no squares outside the board.

Mark seeds with • and added squares with $+$.

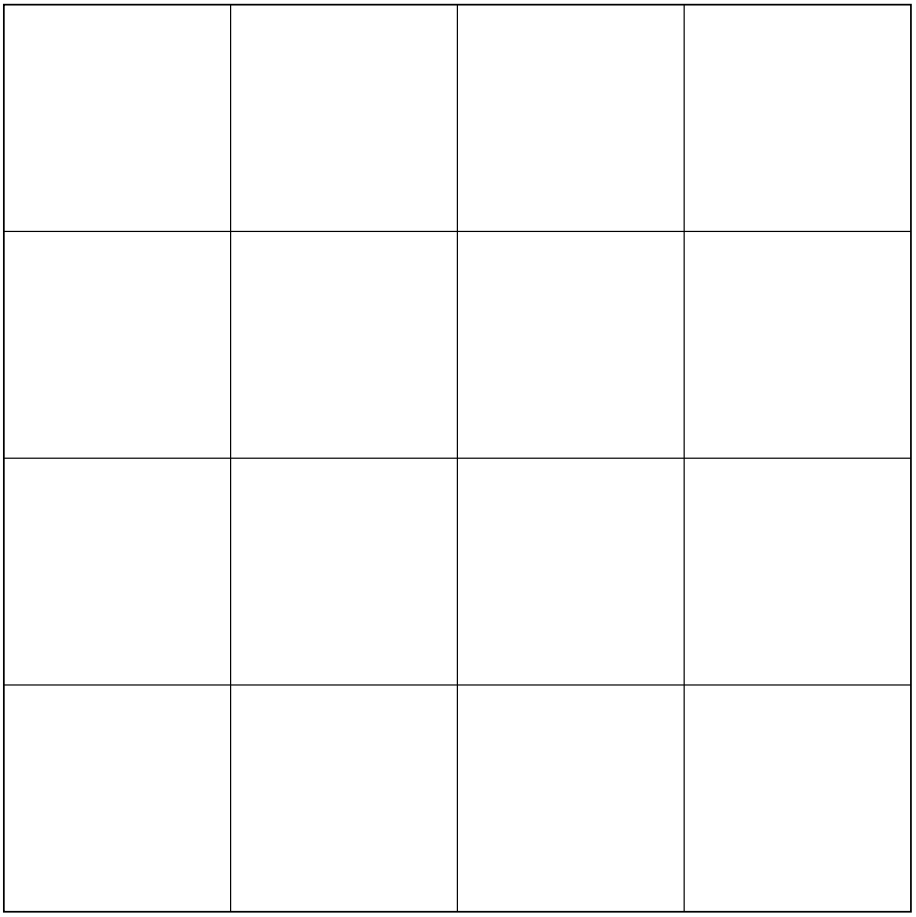


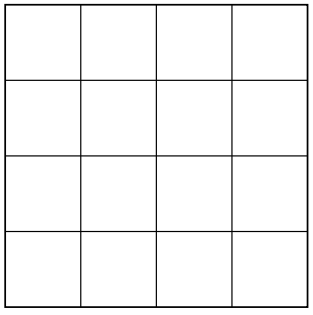
Problem 1: Choose exactly three seeds on a 3-by-3 board. How many different final numbers of filled squares can you make?

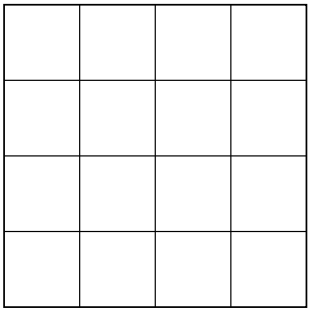


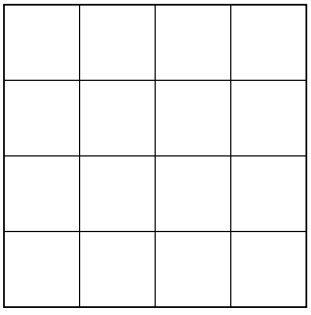


Problem 2: What is the smallest number of seeds you can use to fill a 4-by-4 board? Find three different starts using your best number.

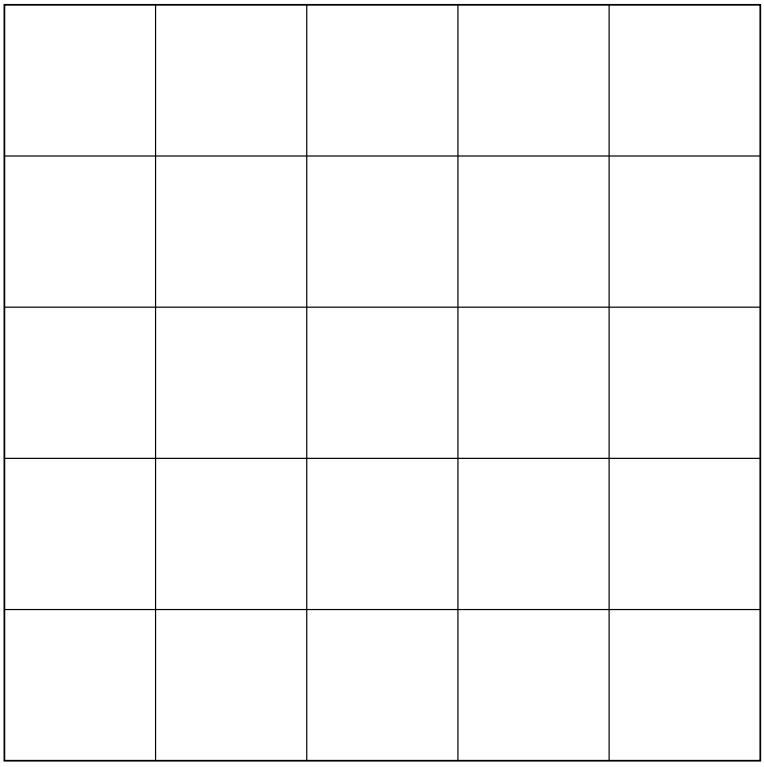


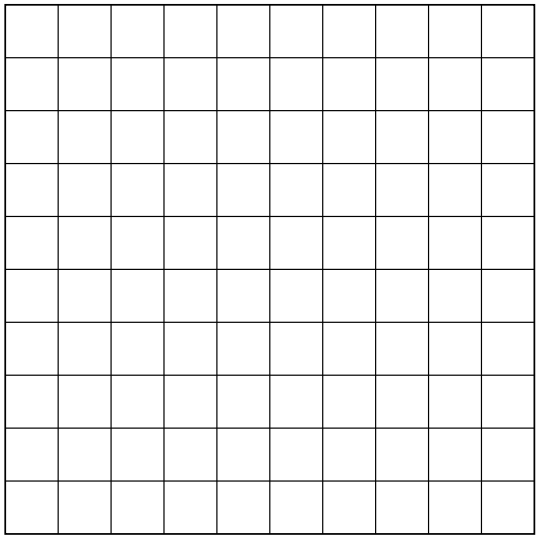




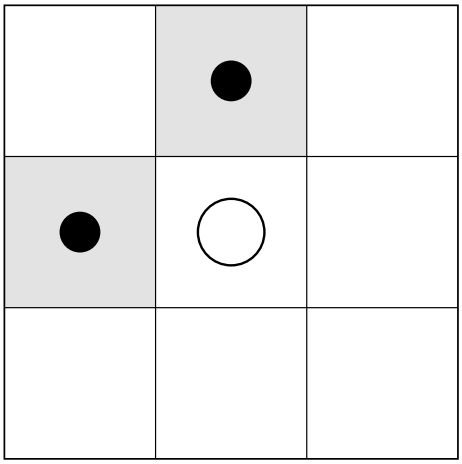


Problem 3: Find the fewest seeds you can use to fill a 5-by-5 board and a 10-by-10 board. Grow these boards by drawing • for seeds and + for additions instead of using counters.



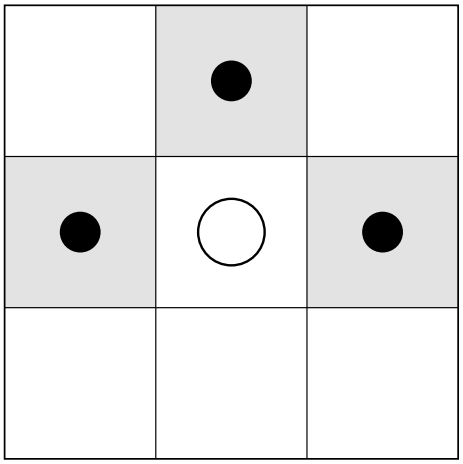


Problem 4: An exposed side is a side of a filled square next to an empty square or the outside of the board. Count the exposed sides before and after filling the circled square in each picture. Can any legal addition increase the number of exposed sides? Explain.



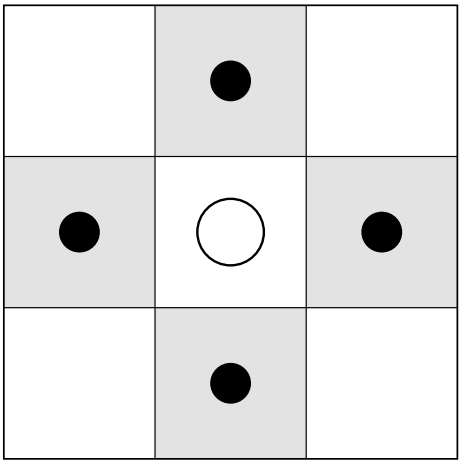
Before: _____

After: _____



Before: _____

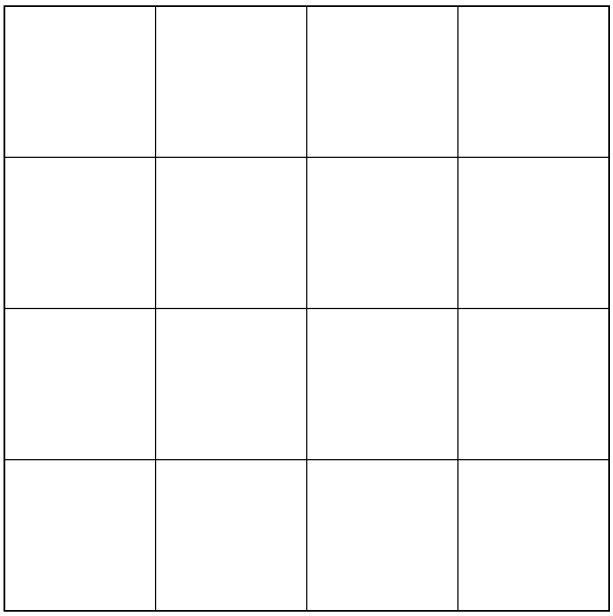
After: _____

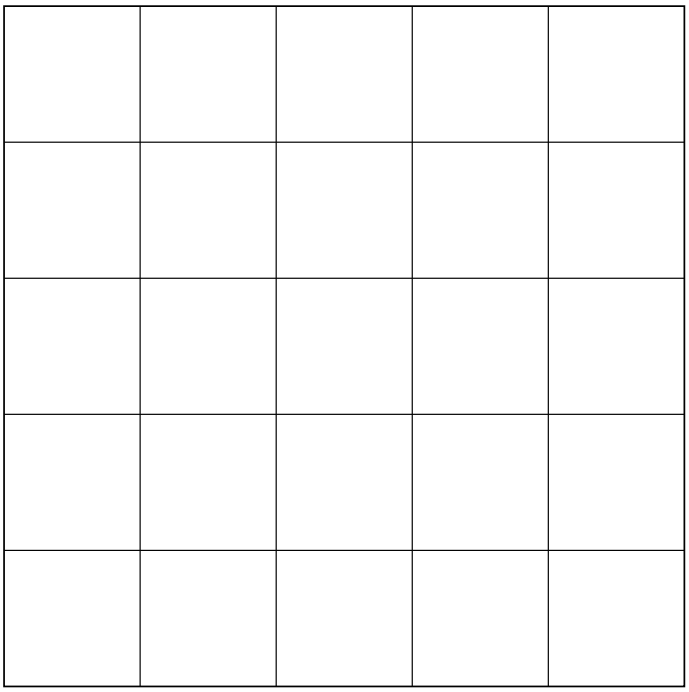


Before: _____

After: _____

Problem 5: Can three seeds fill a 4-by-4 board? Can four seeds fill a 5-by-5 board? Explain why each answer holds for every possible start.





Problem 6: Grow this start, then test all four ways to leave out one seed and start again. Is every seed needed? Is four the smallest number of seeds that can fill the board? Find a different four-seed start that fills the whole board and still needs every seed.

